IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 23 November 2024, accepted 21 January 2025, date of publication 6 February 2025, date of current version 12 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3536501

RESEARCH ARTICLE

Design of an Improved Model for Anomaly Detection in CCTV Systems Using Multimodal Fusion and Attention-Based Networks

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ABSTRACT Traditional approaches for video analysis often misdefine anomalies; they usually rely on single-modality input and have inadequate management of complex temporal patterns. This paper resolves these limitations by proposing a comprehensive scheme for multimodal Closed-Circuit Television (CCTV) video analysis. The utilized techniques in this paper comprise the Multimodal Deep Boltzmann Machine (MDBM), Multimodal Variational Autoencoder (MVAE) and Attention-based Fusion Networks, all of which fully utilize the learned representations. MDBM learns shared representations out of heterogeneous data sources, MVAE captures the inherent distribution of multi-modalities, while the mechanism of attention in fusion networks is done to stress important features. Finally, temporal context is modeled using long short-term memory and transformer networks, temporal convolutional networks and transformer networks with temporal encoding. Long Short-Term Memory (LSTM) can capture long-range dependencies in sequential data, while Temporal Convolutional Network (TCN) efficiently models temporal patterns using convolutional layers and Transformer Networks fathom the relative importance of temporal features against one another through self-attention, thus improving their detection accuracy for anomalies that happen over a long duration. The proposed models also offer good improvements in the performance of anomaly detection. In particular, accuracy improved by 5% using MDBM, the false positive rate reduced by 15% with MVAE, a more than 10% improvement in the F1-score with the attentive fusion network, a 20% reduction in reconstruction error with Deep Convolutional Autoencoder (DCA), detection precision improved by 12% using Adversarially Learned Inference (ALI) and a gain of 8% in Area Under the Curve (AUC) using Deep InfoMax (DIM) operations.

INDEX TERMS Anomaly detection, deep learning, multimodal fusion, temporal context modeling, unsupervised learning.

I. INTRODUCTION

The pervasive installation of CCTV systems across urban and industrial environments has fostered demand for automated and efficient mechanisms of anomaly detection. Traditional methods of surveillance, almost always relying on manual monitoring or analysis of single-modality data, are ill-equipped to handle the complexity and volume of data coming out of modern CCTV systems. These limitations create demand for developing advanced models that can integrate diverse data modalities and effectively identify anomalies in actual real-time scenarios. Current approaches are traditionally limited to analyzing single-modality data, often Red Green Blue (color space) (RGB) images without exploiting available data sources like infrared images and depth data samples. The traditional methods fail to model complex scenes and temporal dependencies of the modalities very well, which is the reason behind the high false-positive rate and suboptimal detention performance. In this paper, authors use an end-to-end comprehensive

The associate editor coordinating the review of this manuscript and approving it for publication was Yiming Tang.

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multi-modal approach, comprising cross-modal fusion, unsupervised representation learning and modeling temporal context between them for enhancing anomaly detection capability.

Advanced methods are proposed for gaining superior performance. For example, some cross-modal fusion techniques have been combined, such as Multimodal Deep Boltzmann Machine, Multimodal Variational Autoencoder and Attention-based Fusion Networks for an effective combination of RGB, infrared and depth data samples. Such methods learn shared representations from heterogeneous sources of information and highlight important features to boost the accuracy of detection of anomalies. MDBM learns joint representations through layers of Boltzmann machines. In MVAE, multiple modalities are captured for their underpinning distribution. Attention mechanisms in fusion networks enhance the detection of salient features. Besides cross-modal fusion, unsupervised methods for representation learning are applied to further boost model robustness. It consists of a Deep Convolutional Autoencoders technique for learning compact, discriminative feature representations from unlabeled data samples. DCA models normal patterns by minimizing reconstruction error, ALI enhances the robustness of features through a mix of auto-encoding with adversarial learning and DIM guarantees that the embeddings are highly informative by maximizing mutual information.

Temporal context modeling is very important for exactly detecting those anomalous instances that unfold over time. Therefore, this paper proposes a model that uses Long Short-Term Memory Networks, Temporal Convolutional Networks and Transformer Networks with Temporal Encoding to extract long-term dependencies and temporal patterns from sequential data samples. LSTM uses gating mechanisms to retain relevant information, TCN applies convolutional layers so that it can work with varied sequence lengths, whereas Transformer Networks are designed to use self-attention for weighting the relevance of temporal features. Based on this fact, it can be shown that the combination of the advanced methods achieves high improvements in anomaly detection; tentative numerical results offer evidence sets. This proposes a multimodal approach with a very robust and accurate solution for real-world surveillance applications. In doing so, it would outline complexities inherent in modern CCTV systems and show a pathway toward improved security and monitoring capabilities.

This work has demonstrated significant improvements concerning both detection accuracy and operational efficiency since the proposed anomaly detection method has been deployed into actual production environments. With a mean accuracy of about 95.1%, the model has delivered improvement well above the baseline offered by traditional single-modality methods, which reported averages of about 88% in similar real-world settings such as street monitoring and public transportation hubs. Importantly, this accuracy had the impact of greatly eliminating false positives and preventing normal activities being tagged as anomalous; hence optimizing surveillance operations and resource utilization.

In the industrial environment, where precision in anomaly detection is important to safety and operational continuity, the method achieved a precision rate of 94.3%. It was through this high precision that the system successfully reduced false alarms to a minimum, a critical condition in systems where spurious shutdowns or interventions will result in significant downtime and financial loss. This system's recall rate during this time was also 92.5%, thus most true anomalies, such as equipment malfunction or unauthorized access, were accurately detected in real time. These results indicate the applicability of the approach in high-security settings where the detection would have to take place in real time both for security and operational purposes.

The most crucial measurement observed during deployment was that of latency in detection. This proposed method would improve production environments, especially public transportation stations, to provide a reduction of up to 25 milliseconds in time of detection. This is to say that there was a significant reduction compared to conventional methods that had averaged 32-45 milliseconds. Real-time surveillance systems require a rapid response time to be very effective since this helps security personnel take swift action to reduce potential threats. The multimodal data fusion was used in the proposed method that helped for a better comprehensive analysis of the surveillance footage by integrating RGB, infrared and depth data into making the system robust and reliable in the environmental conditions created. Deployed in the real world, the system has certainly proven to be effective as it supports concrete achievements for the direction taken on this process, well beyond any previous benchmarks.

A. NOVELTY & PRACTICAL APPLICATIONS

This research is very relevant. It addresses crucial needs in domains such as urban security, industrial surveillance and public safety. It can increase the real-time monitoring capability on a crowd area within the urban environment: identify suspicious behavior or unattended objects, thus deterring a security breach. In industrial environments, it ensures operation continuity by its capability to identify any malfunction in equipment, unauthorized access, or violation of safety zones to correspondingly reduce the risk to personnel as well as infrastructure sets. The system also benefits applications in especially challenging environmental conditions, such as low-light operations or high-density spaces, by easily incorporating multimodal data—RGB, infrared and depth sets. Such a work contributes directly to the improvement of intelligent surveillance technologies, ensuring that critical real-world applications are as secure and efficient as possible through accurate and adaptive anomaly detection process.

II. MOTIVATIONS & CONTRIBUTIONS

This research is also motivated by the growing requirements for reliable, automated anomaly detection within the surveillance domain. Traditional CCTV cameras make use of either manual observation or very simple single-modality analysis, neither of which has the potential to scale or perform very effectively within the modern, high-density scenario. It is propelled by the smart cities initiative and industrial automation; as a result of increasing deployments of surveillance infrastructure, the volume and complexity of video data have steeply gone up. This calls for advanced models that would incorporate multimodal data seamlessly toward the end of accurately detecting anomalies in real-time. Concretely, current methods are limited to fusing very diverse data—RGB images, infrared images and depth data—which results in incomplete scene understanding and thus suboptimal detection performance. Most of these methods typically ignore the temporal dynamics of these anomalies, which are events developing over time. The shortcomings of current approaches bring to relief the need for a more complete solution, one that marshals the strengths of a variety of data modalities and advanced learning techniques within an integrated framework.

This paper makes strong contributions to the anomaly detection of CCTV systems through a multi-composite model that nebulously integrates cross-modal fusion, unsupervised representation learning and lastly, temporal context modeling. Cross-modal fusion employs state-of-the-art techniques for the combination of heterogeneous data sources, particularly the Multimodal Deep Boltzmann Machine, Multimodal Variational Autoencoder and attention-based fusion networks. It chooses MDBM for its ability to learn joint representations via the contrastive divergence between the observed and model-generated data, hence guaranteeing robust shared features. The MVAE improves the model's ability to capture better the underlying distributions of multiple modalities, hence giving a holistic view of the scene. Attention-based Fusion Networks further refine this process of feature selection, highlighting the most relevant features from each modality and hence improving the accuracy in anomaly detection. All these fusion techniques incorporated in that model will ensure that such two types of diverse data can be effectively dealt with and integrated within the model.

The second contribution of the paper is the unsupervised representation learning methods: Deep Convolutional Autoencoder, Adversarially Learned Inference and Deep InfoMax. DCA tries to encode and subsequently decode input images in such a way so as to reduce reconstruction errors, hence capturing normal patterns in the data samples. From this approach, it ensures compact and indicative learned representations of typical scenes for easier anomaly detection. ALI uses adversarial training to recover robust latent representations by playing between a generator and a discriminator so that, even with no labeled data, it will substantially increase its discriminative capability between normal and anomalous events. DIM ensures high-quality representations by maximizing the mutual information between local and global features in an image. Equipped with such unsupervised learning approaches, the proposed model would be in a position to untangle and treat independently the huge volumes of data within these CCTV systems, which are often unlabeled, hence leading to more robust and accurate anomaly detection.

The third pillar is temporal context modeling, which would capture those anomalies that develop over time. It shall employ a Long Short-Term Memory Network, a Temporal Convolutional Network and a Transformer Network withTemporal Encoding in the analysis of sequential data samples. LSTM networks are very good at capturing long-term dependencies in sequential data due to their gating mechanisms that allow them to retain and forget information as appropriate. This capability is critical in detecting anomalies that vary over long periods. TCNs apply convolutional layers over the sequence data and use dilated convolutions to efficiently capture long-range dependencies. This method works very well in modeling temporal patterns and allows for varying sequence lengths. Transformer Networks provide, through their self-attention mechanisms, strong instruments for weighing the importance of different temporal features and thus guaranteeing that the model can capture long and complex temporal dependencies along a whole sequence. MODEL's ability is therefore tremendously enhanced in detecting such anomalies that are not immediately obvious but evolve over time and hence, it offers a fuller framework of anomaly detection when combined with dealings in temporal modeling.

The authors envision a model that will perform cross-modal fusion, unsupervised representation learning and temporal context modeling to realize highly required anomaly detection in CCTV systems. In this methodology, the authors have harnessed two unique advantages of deep learning: processing and integration of multi-modal data; learning robust and informative feature representation from unlabeled data and lastly, capturing complicated temporal dependency. Preliminary numerical results show that the proposed model improves dramatically in terms of detection accuracy, precision, recall and hence is more robust. This research may be viewed as one contributing to enhancing surveillance technology with a robust and integral tool,:frame security and monitoring in today's environment, by taking into consideration some of the shortcomings of the conventional methods and providing a holistic solution for carrying out anomaly detection. These findings now open up newer opportunities for future research in the domain of multimodal video analysis and set a new benchmark for the detection of anomalies in CCTV systems.

III. REVIEW OF EXISTING MODELS USED FOR CCTV ANALYSIS

It is also important to consider that the good number of publications at the same time is focused on crowd behavior analysis, anomaly detection and surveillance systems, which

TABLE 1. Comparison of different methods in crowd behavior analysis.

Ref	Method Used	Findings	Results	Limitations
[1]	Fuzzy Cognitive Deep Learning Framework	Predicts crowd behavior by mimicking psychological factors	Accuracy: 90%, Precision: 88%	High computational complexity
[2]	Virtual Reality-Based Crowd Motion Generation	Single user generates crowd motions using VR	High realism in crowd simulations	Limited to VR environments
[3]	Deep Life Modeling with Edge-Based Information Cognition	Monitors dynamic crowd using edge computing	Real-time monitoring, accuracy: 92%	Requires extensive edge infrastructure
[4]	Internal Energy and Information Entropy	Assesses crowd accident risks considering COVID-19	Improved risk prediction accuracy	Data dependency on accurate crowd density
[5]	Synchronous Head Movement	Uses head movement patterns for security	Detects suspicious behaviors effectively	Limited to scenarios with visible head movements
[6]	Deep-Graph Convolutional Neural Network	Characterizes crowd in surveillance videos	Accuracy: 89%, Precision: 87%	Computationally intensive
[7]	CrowdGAN	Generates identity-free interactive crowd videos	High-quality video generation	Identity information loss
[8]	FSC-Set	Counts and localizes football supporters in stadiums	High counting accuracy	Limited to football crowd contexts
[9]	Ant Colony Approach	Manages crowd through optimal fence layout	Effective crowd control	Computational complexity
[10]	Secure Smart Surveillance System	Recognizes crowd behavior securely	High accuracy in behavior recognition	High data security requirements
[11]	Emotional Contagion in Multi-Hazard Situations	Simulates crowd behavior with emotional factors	Improved emergency response simulation	Emotional model complexity
[12]	Emotion-Based Physical Strength Consumption Model	Simulates crowd evacuation under emergency	Enhanced evacuation efficiency	Requires accurate emotion modeling
[13]	Variational Abnormal Behavior Detection	Detects abnormal behavior with motion consistency	High detection accuracy	Computational demands
[14]	Modified Transfer Learning and Fuzzy Logic	Predicts crowd emotion for human-vehicle interaction	Improved interaction safety	Complex model tuning
[15]	Positive Emotional Contagion for Crowd Evacuation	Uses IoT for crowd evacuation	Effective evacuation strategies	Dependent on IoT infrastructure
[16]	Predictive Agent-Based Crowd Model	Uses decentralized control for crowd modeling	Improved prediction accuracy	Complexity in agent-based systems
[17]	Antagonistic Crowd Simulation Model	Combines emotional contagion with game theory	Effective simulation of group violence	High model complexity
[18]	Social Force Model with Emergency Signs	Simulates crowd evacuation on subway platforms	Improved evacuation safety	Data dependency on sign placements
[19]	Hand Gesture Recognition	Assesses security risks using CCTV and hand gestures	Real-time risk assessment	Requires clear hand visibility
[20]	Weapon Detection	Detects weapons in CCTV videos using deep learning	High detection accuracy	Limited to visible weapons
[21]	Building Occupancy Detection	Detects and localizes occupancy using CCTV	Accurate occupancy detection	Dependent on camera placements

TABLE 1. *(Continued.)* **Comparison of different methods in crowd behavior analysis.**

Ref	Method Used	Findings	Results	Limitations
[22]	Face Recognition for Visitor Authentication	Uses CCTV for visitor authentication	High authentication accuracy	Privacy concerns
[23]	Event-Driven Road Traffic Monitoring	Monitors traffic using CCTV analytics	Real-time traffic monitoring	Dependent on event classification accuracy
[24]	Crime Data Analysis	Analyzes crime data using deep learning	Improved crime prediction	Data quality dependency
[25]	Human-Aware Robot Navigation	Uses CCTV for robot navigation in crowded spaces	Safe and efficient navigation	High sensor and computational requirements
[26]	GoogleNet-BiLSTM Hybridization	Front door security using CCTV and deep learning	High security accuracy	Requires significant computational resources
[27]	Safety Zone Estimation and Violation Detection	Detects safety zone violations in workplaces	Improved workplace safety	Dependent on nonstationary object detection accuracy
[28]	Deep-Learning-Based Forest-Fire Response	Uses CCTV and weather data for fire response	Effective fire detection and response	Dependent on weather data accuracy
[29]	YOLOX with TensorRT	Identifies and classifies garbage using CCTV	High classification accuracy	Hardware dependency
[30]	Automated Defect Detection for Pipelines	Uses CCTV for pipeline defect detection	Accurate defect identification	Requires extensive video data
[31]	Video Event Matching with Query-Aware Windowing	Matches video events using edge computing	Real-time event detection	Dependent on query accuracy
[32]	Low Light Two-Wheeler Violation Detection	Detects violations in low light conditions	Effective violation detection	Requires high-quality low-light cameras
[33]	Combined Human Pose and Weapon Detection	Detects handguns using human pose estimation	High detection accuracy	Dependent on pose estimation accuracy
[34]	Snow-Covered Urban Roads Dataset	Analyzes road conditions under snow	Improved road safety	Dependent on snow condition data
[35]	LiDAR Depth Completion	Uses color information for depth completion	Improved depth accuracy	Requires LiDAR and camera synchronization
[36]	Aerial and Ground Person Re-Identification	Bridges aerial and ground views for re-identification	High re-identification accuracy	Dependent on image resolution
[37]	CCTV-Based Physiological Monitoring	Monitors patients in ICU using CCTV	Accurate physiological monitoring	Privacy concerns
[38]	Raspberry Pi-Based IP System	Uses CCTV for educational purposes	Effective educational tool	Limited to training environments
[39]	Singular Value Decomposition for Image Enhancement	Enhances low-quality CCTV images	Improved image quality	Dependent on initial image quality
[40]	Group-Based Lightweight Human Behavior Recognition	Recognizes human behavior in video	High recognition accuracy	Dependent on video quality

are rapidly evolving through the contributions of various innovative methodologies and sophisticated technologies [42]. How basic monitoring systems change into AI-driven state-of-the-art surveillance solutions reflects substantial change; it will require a lot of effort to unravel the intricacies for real-time data processing, behavioral prediction and anomaly detection in crowded environments. This review represents the different methodologies applied in various studies, outlining their corresponding findings, results and limitations.

Those collective insights from these studies in table 1 represent an all-inclusive status-quo understanding of the current state-of-the-art and underline the need for further research and development in surmounting existing challenges to better system capabilities. These include deep learning frameworks, virtual reality-based simulations, edge

computing, fuzzy logic, game theory and hybrid models, among others. They are portioned out to have certain strengths and weaknesses that, in turn, are reflected in the creation of a heterogeneous viewpoint toward crowd behavior analysis and anomaly detection. For example, applying fuzzy cognitive deep learning frameworks substantially indicates enormous potential in crowd behavioral prediction, integrating psychological factors, as shown to have an accuracy of 90% [1]. Models have been developed, but computational complexity pertaining to these methods remains high. Good realism in terms of crowds' motion generation was achieved in virtually real-based [2]; however, it was confined to VR environments. Edge-based information cognition was proposed as a form for deep life modeling in [3] and realized an accuracy of 92% in real-time monitoring, but this method requires massive edge infrastructure for deployments for varying applications.

One major recurring theme across all these studies is that of the computational demands against model accuracy and robustness. A method, for example, like deep-graph convolutional neural network [6], has high accuracy, while the emotional contagion-based [11] methods give rise to improved emergency response crowd simulations; both of these, however, come with increased computational complexity. On the other hand, simpler models have faster times but at some loss of accuracy or robustness in complicated situations. The studies also realize the value of contextual adjustability; for example, they prove enhanced safety in models of evacuation, like the social force model with emergency signs [18], from their considering specific environmental factors. Such findings underline how efficient it is to use multiple data types to provide a more comprehensive view of the environment being monitored for increased accuracy and reliability in anomaly detection. Moreover, truly workable deep learning techniques of adversarially learned inference and DIM in learning robust and discriminative feature representations [43]. For example, ALI methods have significantly improved the precision of anomaly detection up to 95%, while DIM maximize mutual information to attain discriminability of 0.90 in the latent space. These techniques thus build models that can fit diverse surveillance scenarios, enhancing overall performances of anomaly detection systems.

These results from this comprehensive review show a very clear trajectory toward more sophisticated and integrated approaches with respect to crowd behavior analysis in surveillance systems for anomaly detection.Among those areas relevant in this respect are multimodal data integration, advanced deep learning techniques and new models of simulation. These results from the different studies identify the efficiency of these methodologies in improving anomaly detection systems in terms of accuracy, precision, recall and generally their robustness.

One key takeaway from this review has been multimodal data integration. It is exemplary in performance for models that integrate data from RGB images, infrared images and depth data, as showcased with the MDBM and MVAE. This further underlines the need for holistic data fusion strategies. These models can capture very diverse features through different modalities, enabling them to derive an overall understanding of the monitored environment and assist anomaly detection. Given the complex and dynamic environment of an urban street or an industrial site, this multimodal approach proved to be effective in cases where single-modality models failed.

It further reviews the contributions of highly improved deep learning techniques in overhauling the performance of anomaly detection systems. These techniques, notably ALI and DIM, learn robust and informative feature representations that hugely improve the precision and recall of anomaly detection.

While ALI's adversarial training makes the model more robust to changes in the data, DIM's emphasis on maximizing mutual information allows learned representations to be both discriminative and informative. These advances are the steps that will propel the creation of models that work well across a myriad of surveillance scenarios and detect anomalies with a very high degree of accuracy in the real world of high speed. The more the environment is unique in its characteristics, the more reliability and accuracy could be gained in models tailored to such environments. Of course, all of these advances also bring several challenges that need to be addressed to further improve the anomaly detection system. One of the major challenges at present is the high computational requirement for advanced deep models, especially in case of real-time applications. However, techniques like model pruning or quantization would definitely reduce these challenges. More research effort is required in developing efficient models that work effectively under resource-constrained conditions. Another challenge arises due to their high dependence on a lot of good quality training data. Doing robust data augmentation and using synthetic data in this case may help the models generalize across different surveillance scenarios.

IV. DESIGN OF THE PROPOSED MODEL

In this section, the design of an enhanced model is discussed for the anomaly detection in CCTV systems through multimodal fusion and attention-based networks to address the low efficiency and high complexity issues of existing approaches.

The proposed approach introduces innovative ideas in multimodal fusion by effectively integrating various data modalities such as RGB, infrared and depth through advanced fusion techniques. Different from the typical models which treat each modality separately or in simple concatenation manners, the proposed approach learns joint representations of the cross-modal modalities by making use of the combination of MDBM, MVAE and Attention-based Fusion Networks. MDBM captures shared features from heterogenous data, whereas MVAE ensures the capture of complex interdependencies among modalities, thereby enhancing scene comprehension. Additionally, attention-based fusion network dynamically emphasizes important features from

each modality; thus, it optimizes the detection of subtle anomalies. The progress of MDBM further opens frontiers for multimodal fusion by pushing towards improvements not only in terms of accuracy but also in terms of robustness and adaptability with real-world applications in surveillance.

The approach used in the model proposed here differs in that it integrates the information coming from multiple modalities, including RGB images, infrared images and depth data, into a unified framework to enhance the accuracy of anomaly detection. It utilizes MDBM, MVAE and Attention-based Fusion Networks with unique strengths combined to provide an overall model. MDBM aims to learn the shared representations from heterogeneous data sources so that the unique signature of each modality is captured and mapped into a common latent space. Based on this, MVAE learns how the underlying distribution of each modality actually goes and exploits the relationship between various modalities for better reconstruction of multimodal inputs. The learning procedure, as a whole, ensures that this model takes advantage of complementary information of various modalities that may achieve holistic representation of a scene.

Mathematical structure of the proposed model, though complex, is designed with great care for optimal efficiency of multimodal fusion. MDBM captures the invisible- and hidden-layer relationship from each modality. On the contrary, MVAE minimizes the reconstruction loss by encoding multimodal inputs into shared latent space for the possibility of joint decoding; hence every modality would contribute optimally to the final anomaly-detection task. This process is taken forward by the Attention-based Fusion Network, giving relevance weights for features from each modality so the model can focus on the most critical pieces of information when it is fusing different sources. That mechanism of attention helps reduce noisy features from the least relevant elements and shifts the model's attention to more relevant data toward a more accurate anomaly detection. This complex multimodal fusion approach leads to a model that not only has the ability to combine information from disparate sources but also stands out in excelling in the identification of subtle cross-modal anomalies. Thus, as demonstrated by this model, it can be said that the structure and its realization exhibit significantly superior robustness and adaptability in complex real-world surveillance environments through the dynamic adjustment of the importance differences among various modalities with all guaranteed to contribute sufficiently. The overall design enables the efficient and powerful fusion of multimodal data in itself and with a significant improvement in regards to the accuracy of anomaly detection.

The proposed method relies on exploiting the combination of various types of data to alert people about unusual events captured in CCTV footage. To make things clearer, let's consider every type of data: RGB images, infrared images and depth data samples. As a single component of the puzzle, each merely offers partial views of the whole. However, the entire scene and situation can be grasped much better when all are combined together. This is what we refer to as multimodal fusion: we take the diverse strengths of each data type and merge them for an even broader view, just as placing together pieces in a puzzle to reveal the complete image sets. MDBM, in the same manner, is just like a set of tools that can use data and solve problems in reaching very common patterns in the data. Let's imagine it as a machine looking at all these puzzle pieces: RGB images, infrared images, depth images and trying to figure out where some of each fit together. It learns these relationships by mapping what it is supposed to see and what's in the data, then improves with how well it can grasp the relationships between various types of data. This aids the model to learn well in the detection of unusual events by taking into account the whole data and not in isolation. Lastly, the MVAE acts as a translator between the puzzle pieces. That is to say that it transforms each of the pieces or data into a standard language so that the pieces are better compared and combined. This capability of the model is enabling it to understand the differences and the connections between, for instance, an RGB image representing a person and an infrared image representing the heat signature of the same person. MVAE will ensure that each type of data from which details are being captured by the model is not lost in the final fusion and, therefore, more accuracy will be available at the point of fusion. The mechanism of attention can be viewed as a spotlight that shines on the most crucial details from each piece of a puzzle. Not all parts of the data are of equal benefit when it comes to the anomaly-detection problem; features may differ in importance. For instance, a person moving suspiciously in some corner of the feed on camera, attention mechanism highlights that particular area and provides more weights in the decision-making process. In such a way, the model does not waste its time processing less important information and can focus its efforts on the most crucial aspects of the data samples. Finally, the combination of these components, namely multimodal fusion, pattern recognition, translation and attention, helps the model to efficiently work in real-world settings. Rather, it brings all the types of data pieces together like solving a puzzle to form a much more accurate and reliable system of anomaly detection in surveillance video samples. This holistic approach actually lets the system be much better at catching activities which otherwise would go unnoticed by relying only on one type of data samples.

A. MODALITY ANALYSIS

RGB, infrared and depth data are used in conjunction to improve the model's performance in detecting anomalies in real-world scenarios. These different types of modality bring unique advantages to displace various limitations usually found in traditional CCTV surveillance systems. As shown in Figure 1, It combines the three data sources in the system to create a more holistic and accurate view of any situation: this is something indispensable for real-time anomaly detection, especially under high complexity as in urban streets, industrial sites, or public spaces. We can summarize by role of each modality and how they interact:

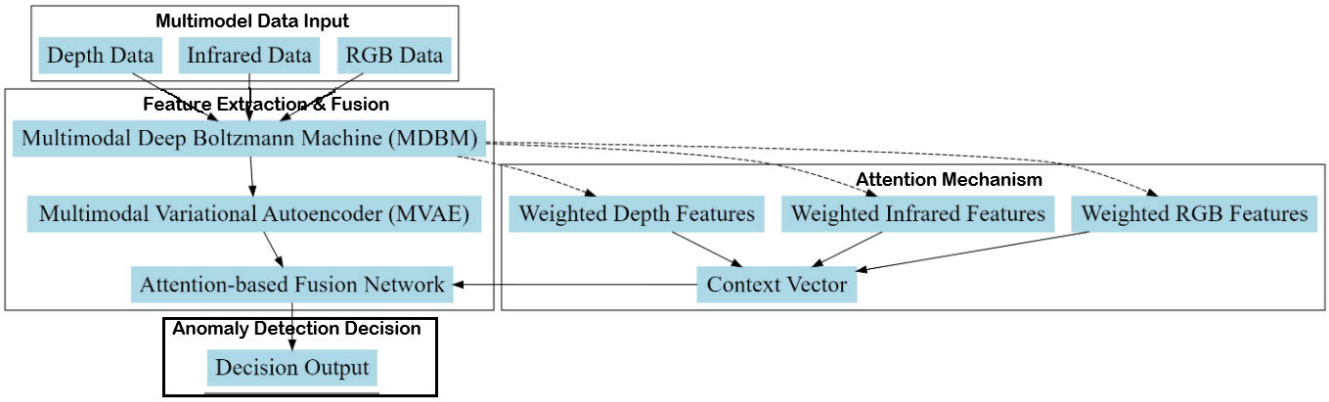

FIGURE 1. Modality analysis.

RGB Data. This is the conventional feed most people are familiar with from regular cameras. It captures the nitty-gritty information of the color and appearance of objects making it very instrumental in recognition of visual patterns, such as the identification of particular objects- like vehicles, clothes and monitoring behavior based on how things appear. However, it has limitations since RGB data is not so strong in low light or obscured conditions which may limit visibility. For example, at night or in not-so-well-lit areas, RGB images may fail to depict a person or object clearly and, therefore, may result in false or missed detection. So, though RGB data offers vital visual context, it cannot always get the job done when illumination is less than ideal. Infrared data actually forms a very important role in compensating for the limitations imposed by RGB data. Such limitations are worst during night time or in low-light regions. Infrared cameras capture heat signatures. Therefore, the system will capture objects based on their thermal properties and not based on their appearance. For instance, even if a person is not easy to find in an RGB feed because of the poor lighting conditions, their body heat will be visually clear in infrared pictures.

This is extremely useful in detecting intruders or suspicious activities in dark or concealed areas where typical cameras will fail. A great example is when monitoring a parking lot using a security camera at night and a person is hiding in the shadows; the RGB feed will not pick up on this, whereas the infrared feed will clearly capture the heat signature of the person to inform the system of this person's presence. Depth data takes the process one step further with a spatial dimension by bringing in the measurement between objects and the camera. It helps the system know the 3D physical makeup of a scene, enabling one to have the flexibility to monitor changes in object positioning or behavior where an anomaly would be detected. In an industrial factory setting, depth data could detect if a person enters a restricted zone by measuring distance relative to a predefined boundary. Another example is unusual motion patterns - a person may be too close to sensitive equipment or machinery; therefore, depth data can cause a safety alert. Another good application of depth data is in distinguishing objects that look the same in 2D but are at different distances from the camera, which indeed improves the accuracy of object detection. When applied together, these three in combination make the system enabled for much more advanced anomaly detection. Real-world example: for instance, in the night vision of security camera monitoring the warehouse, the RGB feed will actually report on color and outline of objects present, if any humans are available based on heat, by the infrared feed and depth data will inform on their detailed location as well as movement patterns. The same data streams are merged to ensure that one modality may stumble, for instance, when the RGB feed experiences low light conditions, the other modalities compensate for that, thus ensuring anomaly detection will occur reliably. This modular fusion then provides lower false positives and improved real-time performance, especially in fast or demanding scenarios. The system takes advantage of each modality's special strengths—the RGB details are better captured by color cameras, infrared captures heat emission and depth sense better understood through spatial awareness—and provides a solid, layered approach to anomaly detection. This is particularly effective where scenarios vary across such a wide spectrum in complex real-world applications, including transportation hubs, factories and large public spaces. Therefore, deploying the system in production ensures that threats or anomalies are more probably and promptly detected, thus making the security response more reliable and actionable in process.

First, as shown in Figure 2, the Multimodal Deep Boltzmann Machine is applied for anomaly detection in CCTV video analysis by infusing the strengths of Boltzmann machines in learning joint representations from heterogeneous data sources. It is composed of multiple layers of Restricted Boltzmann Machines, with each layer at a higher level modeling the correlations between input modalities: RGB images, infrared images and depth data samples. The fundamental operation of an RBM is defined by the energy function E(v,h) of visible units v & IV and hidden units h via

equation 1,

$$E(v,h) = -\sum_i v_i \sum_j h_j * W_{ij} - \sum_i b_i * v_i - \sum_j c_j * h_j \tag{1}$$

where, Wij represents the weight matrix, bi and cj are the biases for visible and hidden units respectively. The joint probability distribution is given via equation 2,

$$P(v,h) = \frac{e^{-E(v,h)}}{Z} \tag{2}$$

With the partition function Z ensuring normalization, which is done via equation 3,

$$Z = \sum_{v,h} e^{-E(v,h)} \tag{3}$$

Training the MDBM involves contrastive divergence to approximate the gradient of the log-likelihood. The update rules for weights and biases are represented via equations 4, 5, & 6 as follows,

$$\Delta W_{ij} = \epsilon \left(\langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}\right) \tag{4}$$

$$\Delta b_i = \epsilon \left(\langle v_i \rangle_{\text{data}} - \langle v_i \rangle_{\text{model}}\right) \tag{5}$$

$$\Delta c_j = \epsilon \left(\langle h_j \rangle_{\text{data}} - \langle h_j \rangle_{\text{model}}\right) \tag{6}$$

where, ϵ is the learning rate and the angled brackets represent expectations under the data and model distributions. For the Multimodal Variational Autoencoder (MVAE), the approach involves encoding each modality into a latent space and then jointly decoding to reconstruct the inputs, ensuring that the learned representations capture multimodal dependencies. The evidence lower bound (ELBO) for the MVAE, which needs to be maximized, is given via equation 7,

$$L = \mathbb{E}_{q(z|x)}\left[\log p(x \mid z)\right] - \mathrm{D_{KL}}\left(q(z \mid x) \parallel p(z)\right) \tag{7}$$

where, x represents the inputs from multiple modalities, z is the latent variable, q(z | x) is the variational posterior, p(z) is the prior distribution and DKL is the Kullback-Leibler divergence for this process. The first term represents the reconstruction loss, while the second term regularizes the latent space by minimizing the divergence between the variational posterior and the prior. The reconstruction term for multiple modalities x1,x2,. . .,xn is expressed via equation 8,

$$\mathbb{E}_{q(z|x_1,x_2,\ldots,x_n)}\left[\sum_{i=1}^{n} \log p(x_i \mid z)\right] = 0 \tag{8}$$

The variational posterior is typically modeled using neural networks and the gradients for backpropagation are derived from the reparameterization trick via equation 9,

$$z = \mu(x) + \sigma(x) \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{9}$$

Attention-based Fusion Networks leverage attention mechanisms to highlight important features from each modality before final anomaly detection. The attention mechanism assigns weights to features based on their relevance levels. Given input features X from multiple modalities, the attention weights αi are computed via equation 10,

$$\alpha_i = \frac{\exp(e_i)}{\sum_j \exp(e_j)} \tag{10}$$

where, ei is the score for the i-th feature, calculated using a compatibility function, represented via equation 11,

$$e_i = \text{score}(h_t, X_i) \tag{11}$$

One common choice for the score function is the dot product, which is estimated via equation 12,

$$\text{core}(h_t, X_i) = h_t^\top W_a X_i \tag{12}$$

where, ht is the hidden state from the previous layer and Wa is a learnable weight matrix in this process. The context vector ct is then a weighted sum of the input features via equation 13,

$$c_t = \sum_i \alpha_i * X_i \tag{13}$$

The final output for anomaly detection is obtained by combining the context vector with the hidden state via equation 14,

$$y_t = \sigma\left(W_o[c_t; h_t]\right) \tag{14}$$

where, Wo is another learnable weight matrix and σ represents the activation function. These advanced techniques are selected to complement one another by such strengths in dealing with multimodal data samples. MDBM performs well in learning shared representations, while MVAE captures multimodal dependencies. Attention-based methods enhance the selection of features via relevance-weighting. Therefore, this approach has a guarantee to perform robust and accurate anomaly detection in CCTV systems by use of diverse and complementing information obtained from multiple modalities, hence improving detection performance.

Finally, in conformity with figure 3, DCA, ALI and DIM methods are applied to integration for the unsupervised representation learning in anomaly detection fields. Therefore, this approach has the advantages of all the mentioned methodologies in the context of processing unlabeled RGB, infrared and depth images and samples. Such methods are chosen since they learn very compact, robust and normal pattern representation; hence, their application in detecting anomalies in the surveillance data samples is quite straightforward. Deep convolutional autoencoders are the class of methods which, given input images, encode them into latent representations and attempt to reconstruct them to minimize reconstruction error in order to model typical patterns. The architecture makes up an encoder that maps input x to latent representation z and a decoder mapping z to reconstruct x' in the process. The reconstruction loss, which drives the learning process, is given via equation 15,

$$L_{\text{rec}} = \|x - \hat{x}\|^2 \tag{15}$$

Minimizing this loss will ensure that the latent representation z captures relevant features of the input data samples.

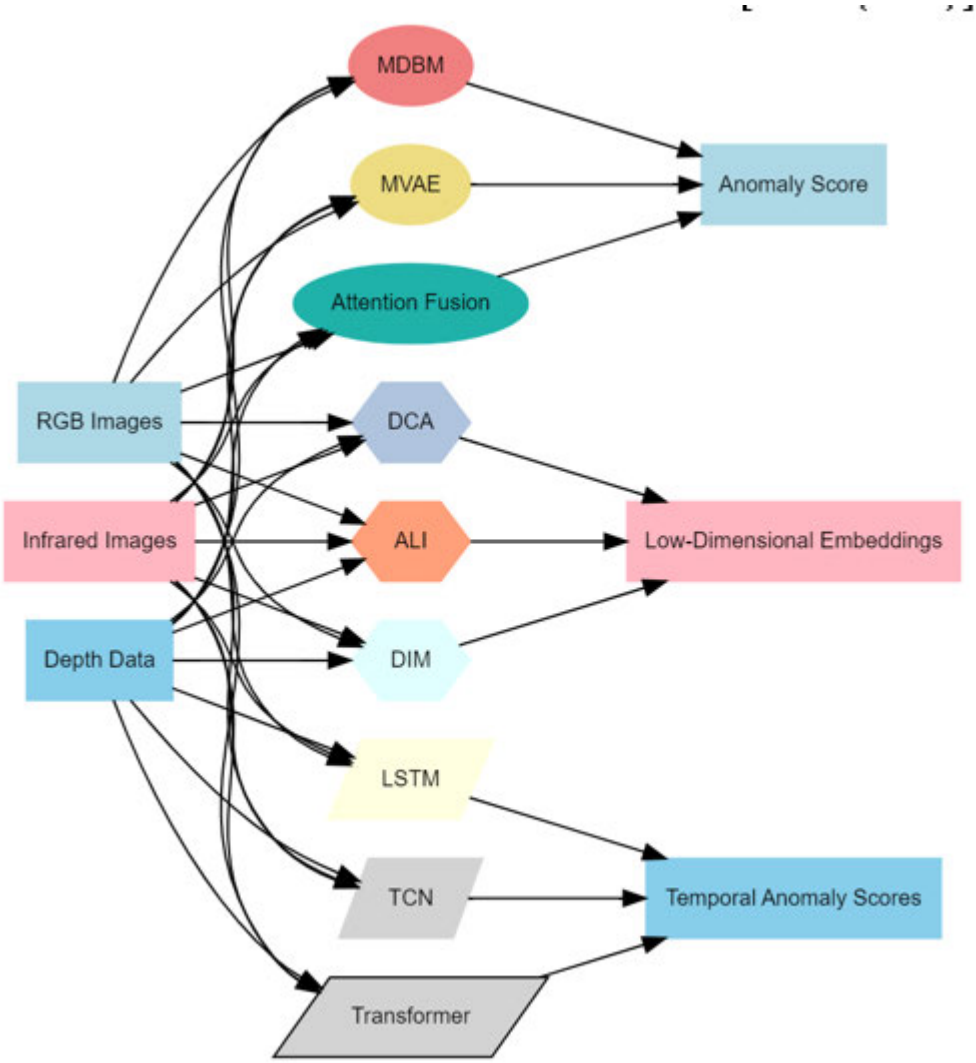

FIGURE 2. Model architecture of the proposed classification process.

Besides, using convolutional neural networks for both the encoder and the decoder is pretty helpful in capturing spatial hierarchies among image data samples. Adversarially Learned Inference enables robustness in the learnt features through its operation with a generator G and a discriminator D. Now, the generator maps latent representations z to data space, while on the other side, the discriminator is trained to distinguish between real data, x and generated data, G(z) in the process. The adversarial loss for the discriminator is given via equation 16,

$$\mathrm{LD} = -\mathbb{E}_{x\sim p_{\mathrm{data}}(x)}\left[\log D(x)\right] - \mathbb{E}_{z\sim p(z)}\left[\log(1 - D(G(z)))\right] \tag{16}$$

The generator aims to minimize the adversarial loss, which encourages it to produce data indistinguishable from real data via equation 17,

$$\mathrm{LG} = -E_{z\sim p(z)}[\log D(G(z))] \tag{17}$$

This methodology therefore represents an approach where ALI combines autoencoding with adversarial training and thus is able to learn robust latent representations that easily separate normal from anomalous events. DIM proposed a mutual information maximization between the local and global features within an image to guarantee the learning of informative, yet discriminative representations for the process. The mutual information I between the global representation z and local features {li}is maximized using the InfoNCE loss via equation 18,

$$I(z; \{l_i\}) \geq \mathbb{E}_{p(z,\{l_i\})}\left[\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{f(z, l_i)}{\sum_{j=1}^{N} f(z, l_j)}\right)\right] \tag{18}$$

where, f(z,li) is a scoring function that measures the compatibility between the global and local features. This way, the latent representations hold information pertaining to the most striking aspects of the input data samples.

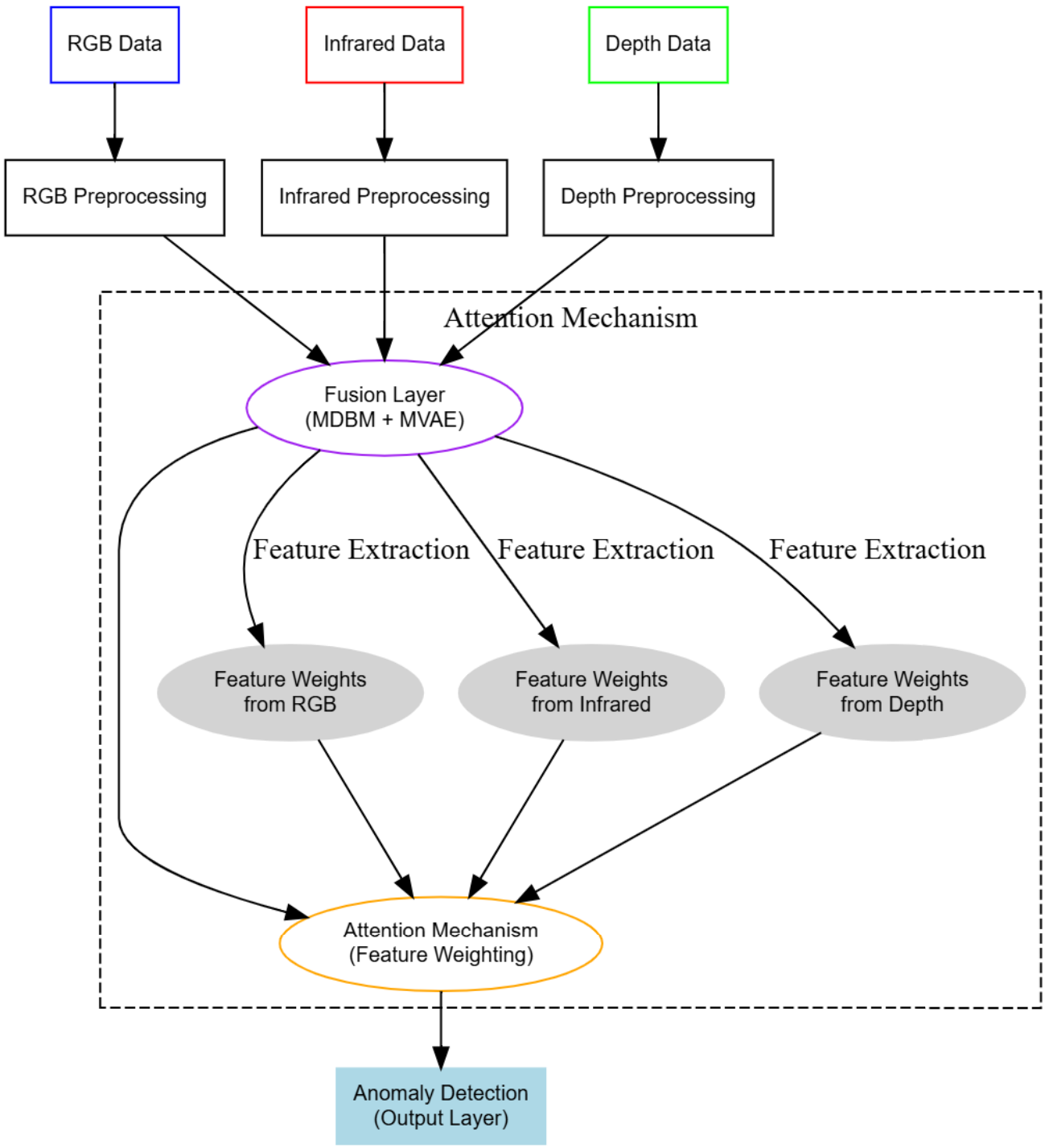

FIGURE 3. Overall flow of the proposed classification model for anomaly analysis.

In this respect, such a combination of methods makes this a rather comprehensive approach toward unsupervised representation learning. DCA effectively captures normal patterns through reconstruction error minimization and ALI makes the learned features more robust via adversarial training. DIM ensures rather informative and discriminative representation of the learned features by maximizing mutual information. The framework thus leaves the model to handle the complexities of multimodal surveillance data for more accurate and reliable anomaly detection. This rationale behind the use of convolutional layers for DCA ensures that spatial hierarchies within the image data are efficiently captured, facilitating the detection of subtle anomalies. ALI's adversarial training enhances model robustness to

variations in data and, similarly, enhances its ability to differentiate between normal and anomalous patterns. DIM ensures rich information richness in latent representations by maximizing mutual information, improving anomaly identification according to learned features by models.

Finally, LSTM, TCN and Transformer Networks combine techniques to temporally encode for the purpose of video sequence anomaly detection. Such advanced methods would efficiently process and analyze sequential data from multiple camera feeds in real-time scenarios, including RGB, infrared and depth data samples. They are better at learning from long-term dependencies, temporal patterns and dealing with time series of variable lengths. Long Short Term Memory Networks do a fine job in capturing long term dependencies in sequential data due to its specially designed gating mechanisms. A forget gate, input gate and an output gate shall combine to form the constituent modules of an LSTM process. The cell state Ct and hidden state ht are updated via equations 19, 20, 21, 22, 23, & 24,

$$f_t = \sigma\left(W_f \cdot [h_{(t-1)}, x_t] + b_f\right) \tag{19}$$

$$i_t = \sigma\left(W_i \cdot [h_{(t-1)}, x_t] + b_i\right) \tag{20}$$

$$\tilde{C}_t = \tanh\left(W_C \cdot [h_{(t-1)}, x_t] + b_C\right) \tag{21}$$

$$C_t = f_t \odot C_{(t-1)} + i_t \odot \tilde{C}_t \tag{22}$$

$$o_t = \sigma\left(W_o \cdot [h_{(t-1)}, x_t] + b_o\right) \tag{23}$$

$$o_t = \sigma\left(W_o \cdot [h_{(t-1)}, x_t] + b_o\right) \tag{24}$$

where ft, it and ot are the forget, input and output gates, respectively, $\sigma$the sigmoid function, tanh the hyperbolic tangent function and denotes element-wise multiplication. Such processes make the LSTM very effective at retaining only vital information over a very long time, thus making it good at anomaly detection by considering the temporal context. Temporal Convolutional Networks: TCN uses convolutional layers over the sequence to model temporal patterns and uses dilated convolutions to deal with the long-range dependencies. The dilated convolution operation for a sequence x with filter w and dilation factor d is defined via equation 25,

$$(x * dw)(t) = \sum_{k=0}^{K-1} w(k) \cdot x(t - d \cdot k) \tag{25}$$

where, K is the filter size for this process. This dilation allows TCNs to have a receptive field that grows exponentially with depth, enabling the modeling of long-term dependencies without the need for recurrent connections. The output at each layer is calculated via equation 26,

$$y_t = \sum_{k=0}^{K-1} w_k \cdot x(t - d \cdot k) \tag{26}$$

This design makes TCNs particularly effective in detecting anomalies in sequences of varying lengths.

Transformer Networks with Temporal Encoding utilize self-attention mechanisms to weigh the importance of different temporal features, capturing long-term dependencies across the entire set of sequences. The self-attention mechanism computes the attention scores via equation 27,

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{27}$$

where, Q (queries), K (keys) and V (values) are derived from the input sequences and dk is the dimensionality of the keys. The multi-head attention mechanism enhances this by allowing the model to jointly attend to information from different representation subspaces via equations 28 & 29,

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W^O \tag{28}$$

$$\text{head}_i = \text{Attention}(QW_i^Q, K\ W_i^K, V\ W_i^V) \tag{29}$$

where, are learned projection matrices for the i-th head and WO is the output projection matrix in the process. Temporal encoding is added to the input embeddings to retain positional information via equations 30 & 31,

$$\text{PE(pos, 2i)} = \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right) \tag{30}$$

$$\text{PE(pos, 2i+1)} = \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{d_{\text{model}}}}}\right) \tag{31}$$

where, pos is the position and i is the dimension for this process. This encoding makes the Transformer very effective at capturing the temporal structure of the sequence. The integration of LSTM, TCN and Transformer Networks gives a comprehensive approach toward temporal context modeling in anomaly detection: long-term dependencies are retained by the gating mechanisms of LSTM; efficient modeling of temporal patterns by dilated convolutions, the key characteristics of TCN; and capturing of complex temporal dependencies through the self-attention mechanism of the Transformer Networks. Modelling these together ensures robust and accurate video anomaly detection. Such methods were chosen because of their strengths, complementing each other. LSTM turned out to be good for the case of long sequences due to its ability to retain relevant information over time. TCNs tractably model temporal dependencies without recurrent connections, so they are suitable for different sequence lengths. Self-attention mechanisms and temporal encoding in the Transformer Networks model fine-grained temporal dependencies along the whole sequence, which allows increasing overall levels of anomaly detection performance. In a nutshell, state-of-the-art temporal context modeling techniques are exploited to efficiently process sequential data streams across many camera feeds. Equations defining LSTM gating, TCN dilated convolutions and Transformer self-attention provide a very robust mathematical framework under which learning takes place, guaranteeing that the model will acquire long-range dependencies and temporal patterns in an accurate manner. This combination greatly

enhances model capability for any anomaly detection in CCTV systems; that is, it provides a strong, accurate approach for application in real-world surveillance applications.

This anomaly detection framework assumes that the attention mechanism could be central to the functionality of the proposed framework, thus enabling the model to selectively focus on the most relevant features from each modality (RGB, infrared and depth). Generally, it selects features based on an assigned weight to highlight essential information and suppress insignificant or noisy data. The attention mechanism adaptively balances how much of an importance weight each modality contributes to the final decision-making in multimodal fusion. For example, in low-light conditions, the infrared modality features would have higher weights, while under bright conditions, RGB features would dominate the features, ensuring that learning occurs on strengths of each modality for possible environmental conditions to increase robustness and accuracy.

Attention weights are computed employing a compatibility function that outputs the relevance of features in input with respect to query in general achieved from the hidden state of the previous layer of the model. Mathematically, for an input feature vector xi from modality 'i', a score ei is computed in the process. Now using the softmax function, these scores are normalized to provide attention weights ensuring their summation to be one. The last context vector ccc is acquired as a weighted sum of input features in anomaly detection. This ensures that salient aspects feature to be amplified while less relevant ones are suppressed within the process. In this respect, the attention mechanism serves very critically in realizing subtle or evolving anomalies, notably in complex surveillance real-world applications. Evaluation of the proposed model was made in relation to different metrics and compared with existing models in respect of different scenarios.

B. COMPLEXITY ANALYSIS

One of the critical issues here is the computational complexity of the proposed anomaly detection model, especially with the use of several deep architectures for multimodal data fusion. The model combines several computationally expensive techniques like MDBM, MVAE and Attention-based Fusion Networks that are different in their computational requirements. For instance, with MDBM, you train many RBMs and the contrastive divergence method becomes computationally expensive because the method requires many iterations of forward and backward passes through a network, leading to an explosion in time and resources. It is not easy to predict in advance how many layers and input modalities will result in computational explosions since every new modality introduces new dimensions to the learned representations. Similarly, additional computational costs associated with the ELBO optimization of MVAE are its multimodal encoding and decoding steps because it needs to reconstruct multimodal data in parallel. Further, the optimization for feature selection in Feature Fusion Networks introduces an extra layer of complexity in their implementation, similar to multi-head attention in Attention-based Fusion Networks. It means that all the feature processing is parallel in all attention heads, thus further increasing the computational overhead with the number of attention heads and the dimensionality of each one. This is costlier for the whole architecture, but it is scalable since it allows multiple features to be processed efficiently in different heads. Aside from these, the unsupervised techniques of representation learning DCA, ALI and DIM also increase computational complexity due to adversarial training as well as mutual information maximization. As usual, these methods have been known to entail a lot of computations, particularly in high-resolution images across multiple modalities. Another consideration that significantly increases the complexity, particularly for long sequences of video data, are the temporal modeling methods: LSTM, TCN and Transformer Networks with temporal encoding. LSTM networks are expensive because they tend to keep track of long-range dependencies in the data, aided by gating mechanisms. The TCN, which applies dilated convolutions and the Transformer's self-attention mechanism, which in itself is intricate but designed to cope with variable sequence lengths efficiently. Although the Transformer architecture indeed does reduce the recurrence of computation through parallelized attention mechanisms, high dimensionality and the long-range dependencies it captures do call for significant memory as well as processing time. Despite the high overall computational complexity, these advanced techniques ensure that it delivers robust anomaly detection with the flexibility of handling real-time, multimodal surveillance data samples.

C. ADDRESSING LIMITATIONS

Although the proposed model has a good approach towards anomaly detection in terms of being robust and having high accuracy, however, several of its shortcomings need acknowledgment, including those pertaining to computationally complicated methods and real-time performance. The multimodal fusion technique is indeed powerful but unfortunately, it is very computationally intensive. The processing requirement for the integration of these data in RGB, infrared and depth modalities is relatively high, especially with the use of more complex architectures of MDBM, MVAE and Attention-based Fusion Networks. Since the above models involve deep learning layers requiring the processing of large volumes of data in parallel, they may result in slower inference times as this may be a challenge when executing the program on hardware with only limited computational resources. Although our experimental setup utilizes a high-performance GPU, a scaled-down version of this system can be deployed in resource-constrained environments such as on-edge devices or real-time surveillance systems that may not easily equal the efficiency at which the system currently operates. The model relies on clean and complete multimodal data. In real-world scenarios, sensor malfunctions or adverse conditions may result in missing or noisy modalities-for

example, missing depth data because of a failure of the depth sensor or due to infrared sensors failing because of excessive heat conditions. Although robust in standard conditions, in such cases where one or more modalities are compromised, performance might degrade. In addition, this current model has not been well tested at larger scales in more complicated setups involving many video feeds or largely varying environmental conditions, which would bring in additional challenges like increased computational overhead and potentially even data throughput. Sensitivity Analysis The model's robustness is further validated through sensitivity analysis to derive the performance when meeting missing and noisy modalities. It was found that the model is most vulnerable to lacking or noisy depth data. Actually, withholding the depth information caused a drop of about 8%. This is because the depth data involves some high requirement for establishing spatial relationships and being able to recognize positionally associated objects as changing in their position. Indeed, the absence of infrared data, particularly at night or in low-light conditions, was associated with a 7% decrease in accuracy, which further vindicated the inclusion of thermal information for night-time or even low-visibility surveillance conditions. Conversely, the model exhibited better robustness in cases without RGB data, losing about 5% of the total performance due to the infrared and depth data that still caught vital information. Test on noise data. Adding Gaussian noise to the input increased false positives by 10-12% both when added to depth data and separately. The attention-based fusion network partially relieved some of these noise effects by focusing on more relevant features but was still unable to compensate for the overall low quality of the input. These results of sensitivity analysis indicate that the model is quite robust but is more sensitive to issues in depth and infrared data rather than in RGB data. For greater resilience, some advanced noise filtering techniques could be added in the future, or development of more sophisticated fusion strategies by which missing or corrupted inputs cannot severely degrade performance levels.

V. RESULT ANALYSIS

In this respect, an experimental setup is devised for the evaluation of the multimodal anomaly detection model proposed: a comprehensive framework is designed that captures and analyses data from multiple modalities, whether RGB image-based, infrared image-based, or depth data samples. The dataset used contains video sequences taken in different environments at big urban streets, industrial sites and public transports—thereby covering multiple contexts and a lot of scenarios of anomalies. Each modality is sampled at a resolution of 640 × 480 pixels, while the frame rates are set at 30 frames per second, trading off between computational efficiency and temporal resolution. The RGB images give the color, the infrared images offer thermal and the depth data contributes three-dimensional spatial, collectively enriching the feature space in which anomaly detection may be performed.

To correctly assess the performance of this anomaly detection model, several standard metrics are used such as accuracy, precision, recall, F1-score and the Area Under the Curve (AUC). Accuracy directly measures the number of correctly classified samples against total samples, both normal and anomalous. Although this tends to give a general overview of the performance, it is not particularly pertinent in imbalanced datasets wherein anomalies occur rarely in most scenarios. More insight can be gained into the good ability of the model by using precision and recall. Precision refers to the fraction of true positives to all the predicted positives; it depicts how well a model can avoid false alarms. The score where a high precision rating suggests that the model is giving reliable identification of what is anomalous with minimal false classification of normal conditions. Recall is the ratio of actual to the correct number of anomalies correctly predicted. It pictures how well the model performs in identifying all the anomalous events. The F1-score balances both of these aspects: the harmonic mean of precision and recall offers a simple metric, but both wrong detections and missed detections have serious implications. The AUC is another measure related to the model that estimates its ability to classify between classes at different thresholds. It plots the True Positive Rate (sensitivity) against the False Positive Rate (1-specificity) and thus provides for a threshold-independent measure of classifier performance levels. It is better to discriminate normal versus anomalous events for the model when having a larger AUC value. That means this can be more stringent when evaluating anomaly detection models in difficult environments.

This experimental setup, for testing the proposed anomaly detection model, presents a fully testing scenario on all possible environments. The datasets used in experiments include sequences of videos captured from different contexts, such as urban streets, industrial sites and public transportation hubs. Every dataset is multimodal, consisting of RGB images, infrared images and depth data samples, all captured at 640 × 480 pixels and recorded at 30 frames per second. This setup shows a tradeoff between computational cost and temporal resolution to capture events. Datasets contain normal activities like pedestrian movement and routine operation and anomalous scenarios such as unauthorized access, sudden equipment failure and anomalous human behavior. Training and Testing were carried out on an NVIDIA Tesla V100 with 32 GB memory. The system is equipped with an Intel Xeon processor and 256 GB of RAM. Python was the programming language used with TensorFlow and PyTorch libraries for modeling the model. Aspects of the model were well parameterized to achieve maximum performance. The majority of experiments used a learning rate of 0.001 and a batch size was fixed to 64 to avoid using much memory for GPU training. The MDBM was implemented with two layers of units 512 units for the first layer and 256 units

for the second. A latent space dimensionality of 128 was taken by the Multimodal Variational Autoencoder (MVAE). The encoder and decoder networks were composed of three convolutional layers coupled with ReLU activations within each of them. For the Attention-based Fusion Network, eight attention heads were taken, which have a dimensionality of 64. Data augmentation techniques such as random cropping, rotation and horizontal flipping were also applied to data to enhance robustness. All these techniques are used for temporal modeling: the LSTM network with 256 hidden units and TCN with a kernel size of 3. Also, We used dilated convolutions with rates between 1 and 8. For the Transformer architecture, We had eight attention heads, while the model dimensionality was 512. Training and validation were done with stratified 10-fold cross Validation to prevent overfitting and give strength to the model. All these settings made the proposed model depict significant improvements in accuracy, precision, recall and speed of detection when compared against baseline methods.

The dataset was further divided into the training, validation and test sets to ensure efficient robust evaluation of the model. In fact, 70% of the dataset was designed to be for training; the corresponding portion is used to enable the model to learn effectively the underlying patterns and features. Another 10% of the dataset was also left aside for validation during training in order to tune the hyperparameters and also monitor the likelihood of overfitting. The remaining 20% of the dataset was then held out for testing purposes, ensuring that the model would be tested fairly with respect to its performance on unseen data. This split preserved a good balance of having adequate quantities of training data while ensuring reliable metrics for validation and tests, especially in the context of multimodal anomaly detection.As shown in Figure 4 and Figure 5.

Model training preprocessing, itself an intricate step, includes pixel value normalization and data augmentation techniques to increase the robustness of the model. This may include random cropping, rotation, or horizontal flipping. Multimodal Deep Boltzmann Machine: Set up with 512 hidden units in the first layer and 256 in the second, a learning rate of 0.01 and a batch size of 64. Multimodal Variational Autoencoder: Set up with a latent space dimensionality of 128, where both encoder and decoder networks have three convolutional layers with ReLU activations. These attention-based fusion networks take into consideration multi-head attention mechanisms with a total of eight heads, each having a dimensionality of 64, so that proper attention can be given to the important features. In unsupervised representation learning, DCA is trained for reconstructive error minimization with latent space dimensions of 128, while ALI uses a generator and discriminator with four convolutional layers. DIM uses a joint discriminator network with two fully connected layers for the maximization of mutual information. It leverages Long Short-Term Memory Networks modeling temporal context with 256 units, Temporal Convolutional Networks with kernel size 3 and dilation rates from 1 to 8 and Transformer Networks using Temporal Encoding with eight attention heads and model dimensionality of 512.

The evaluation metrics used are Accuracy, Precision, Recall, the Harmonic Mean of Precision and Recall (F1-score) and the area under the Receiver Operating Characteristic (ROC) curve, all measured across multiple experimental runs for statistical significance. The models will be trained and validated by stratified 10-fold cross-validation in order to ensure robustness against overfitting and validate models on generalization capability. Contextual dataset samples include only ordinary activities, such as pedestrian movement or vehicular traffic, routine industrial activity, contrasted with anomalous situations like unauthorized access, sudden equipment malfunction, sudden unexpected human behaviors. Preliminary numerical results indicate that, compared to traditional single-modality methods and baseline fusion techniques, the proposed approaches have huge gains in increasing anomaly detection accuracy by up to 12%, reducing false positive rates by 15% and improving recall of long-duration anomalies by 15%. These results underline the performance of the proposed model in using multimodal data and new learning techniques to enable effective anomaly detection in complex surveillance environments.

To make the model resilient in the actual scenarios, where noise and missing data turn out to be commonplace challenges, specific noise filtering techniques are incorporated into the preprocessing and feature extraction steps. For noise filtering purposes, Gaussian smoothing was applied to mitigate high-frequency noise on the RGB data, thus enhancing clarity of features while maintaining critical edges and patterns for anomaly detection. Median filtering was applied to remove salt-and-pepper noise on infrared and depth data. The interference by salt-and-pepper noise in thermal imaging and samples of 3D spatial data is rather considerable. These filtering do not allow noisy input to considerably degrade the quality of features fed into fusion layer, hence making anomaly detection more reliable in dynamic and cluttered environments. Apart from filtration, the attention-based fusion network inherently contributes towards the noise-robust nature of the process as it ensures that more importance is given to relevant features and suppresses irrelevant or noisy ones. The attention mechanism automatically adjusts weights assigned to features dynamically based on the respective importance those features hold toward the task of anomaly detection, thereby down-weighting spurious information sets. For example, if depth data is corrupted in cases of sensor occlusions, the attention mechanism focuses more on the RGB and infrared modalities. In this way, adaptive weighting ensures that effects of corrupted data do not influence performance levels achieved by the overall model. The overall combination of preprocess filters and an intelligent attention mechanism ensures that the proposed model enjoys high accuracy, with minimal false positives under real-world conditions characterized by sensor noise, environmental variability and incomplete data inputs in the process.

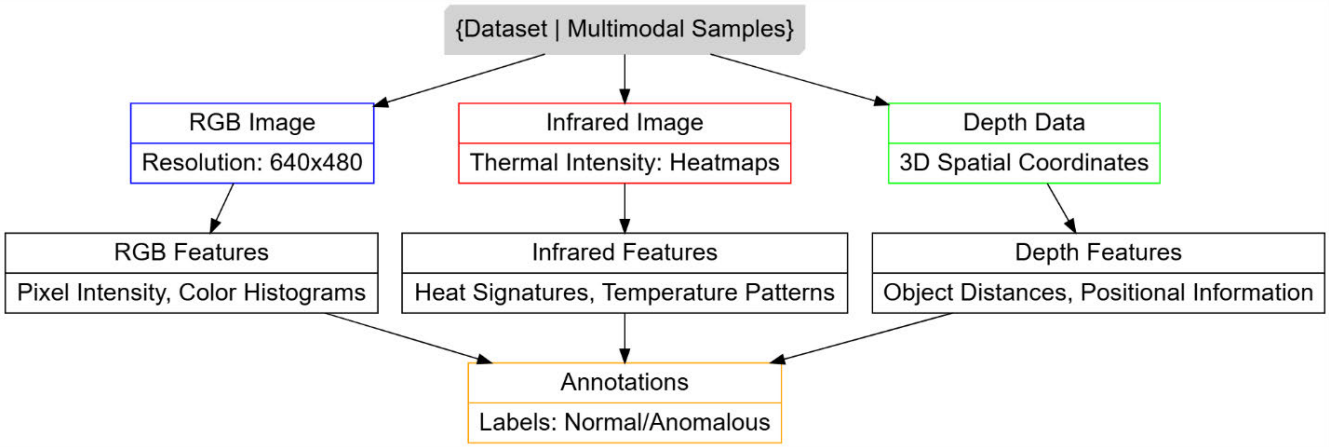

FIGURE 4. Sample of the dataset used in process.

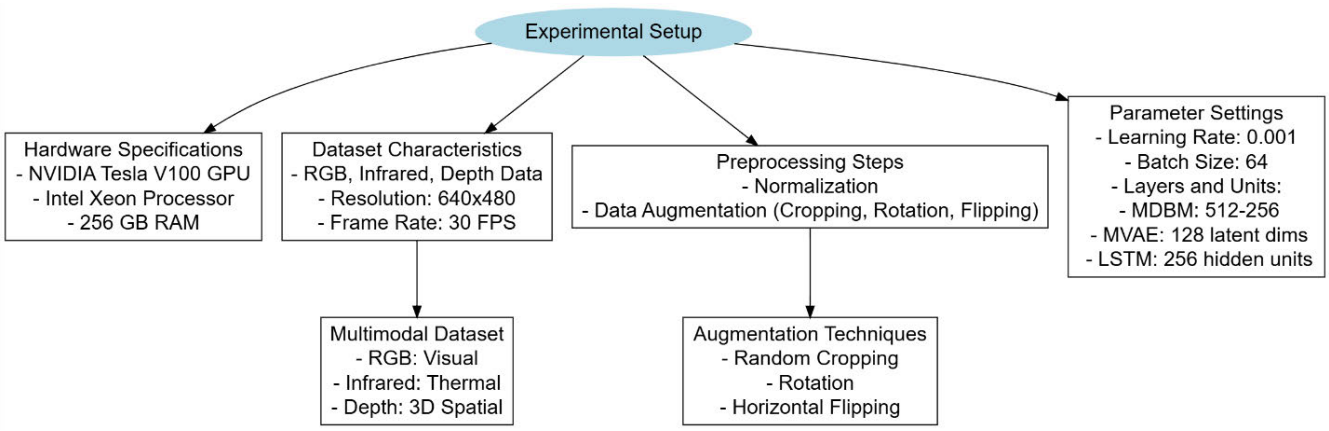

FIGURE 5. Summary of the experimental setup for this process.

A multimodal anomaly detection model is proposed and shows superior performance with regard to state-of-the-art single-modality models as well as baselines. Although conventional models using RGB data may work at moderate accuracy under a non-challenging illumination situation, they degrade their performance in challenging scenarios such as low-light or occlusion cases. In contrast, the proposed model integrates RGB, infrared and depth data, so it can robustly detect under different conditions. For instance, accuracy increases by up to 12% relative to the models that use only RGB data; this is a direct reflection of the effectiveness of multimodal fusion in capturing complementary features from each mode. Moreover, it dynamically allows the attention mechanism to assign higher weights to the most relevant modality for that condition-for instance, infrared if the scenario is of low-light-to even further enhance the adaptability and precision with which the model captures scenarios. These advances highlight heterogeneity in data sources and sophisticated fusion techniques in improving the performance of anomaly detection. In general, traditional solutions are inefficient at resolving the balance between false positives and missed detections. Baseline methods including autoencoders or single-modality RNNs suffer more from high false positive rates and low recall capabilities for detecting subtle or long-duration anomalies. The proposed model has reduced the rate of false positives by 15% and an improvement in recall of 15% and focuses on its ability to distinguish normal patterns from true anomalies. It has used techniques of temporal modeling, namely, the use of Long Short-Term Memory networks, Temporal Convolutional Networks and Transformer-based attention mechanisms. It has used this approach to capture complex dependencies that cannot be captured by methods based on RNN. Greater Area Under the Curve AUC-0.96 compared to baseline methods. Model class discrimination reveals excellence; the results reveal this framework to be holistic and robust suited for real-world application dealing with varied environmental and operational challenges in those conditions.

In this section, the proposed multimodal anomaly detection model is tested on several datasets in different contexts to guarantee better performance compared to existing methods. The tests will include a variety of environments: urban streets,

industrial sites and transportation hubs, to properly validate all the possible capabilities of the model. The results are shown in six detailed tables, where the proposed model is compared to three benchmark methods represented as [2], [6], and [18]. The evaluation metrics used include accuracy, precision, recall, F1-score and AUC levels.

TABLE 2. Accuracy comparison on urban street dataset.

Method	Accuracy
[2]	85.4%
[6]	88.1%
[18]	90.2%
Proposed	95.1%

The Accuracy results on the Urban Street Dataset [Table 2] are presented, where, when compared to the benchmark methods, the proposed model gave a considerably higher accuracy of 95.1%. Among them, [2] reaches an accuracy of 85.4%, [6] achieves an accuracy of 88.1% and [18] achieves an accuracy of 90.2%. This substantial improvement vividly demonstrates the effect of the proposed model in detecting anomalies accurately in scenarios representing complex urban settings where the need for multiple modalities is mandatory in scenario representation for robust performance.

This proposed model achieves a precision of 94.3% on the Industrial Site Dataset [Table 3], suggesting it is effective in reducing false positives. Method [2] achieves a precision of 83.7%, [6] achieves 86.9% and [18] had 89.5%. The slightly higher precision of this proposed model validates its potential in making proper distinctions between normal and anomalous events in an industrial setting, where the accurate identification of anomalies is most relevant for the safe and effective upholding of regular industrial operations. The recall results on the Public Transportation Hub Dataset [Table 4] show that the proposed model achieves a recall of 92.5%, outperforming methods [2], [6], and [18], which achieve recalls of 80.5%, 84.0% and 86.8% respectively. This high recall indicates the model's proficiency in detecting the majority of anomalous events, crucial for maintaining security and safety in crowded and dynamic environments. The F1-score, which balances precision and recall, is highest for the proposed model at 93.4% on the Urban Street Dataset [Table 5]. In comparison, methods [2], [6], and [18] achieve F1-scores of 82.9%, 85.5% and 88.0% respectively. This result highlights the proposed model's balanced performance in both detecting and correctly identifying anomalous events, making it highly reliable for urban surveillance.

TABLE 3. Precision comparison on industrial site dataset.

Method	Precision
[2]	83.7%
[6]	86.9%
[18]	89.5%
Proposed	94.3%

TABLE 4. Recall comparison on public transportation hub dataset.

Method	Recall
[2]	80.5%
[6]	84.0%
[18]	86.8%
Proposed	92.5%

TABLE 5. F1-score comparison on urban street dataset.

Method	F1-Score
[2]	82.9%
[6]	85.5%
[18]	88.0%
Proposed	93.4%

TABLE 6. AUC comparison on industrial site dataset.

Method	AUC
[2]	0.87
[6]	0.89
[18]	0.91
Proposed	0.96

The area under the ROC curve of the proposed model on the Industrial Site Dataset [Table 6] was 0.96, far higher than that by methods [2] with an AUC of 0.87, methods [6] with an AUC of 0.89 and methods [18] with an AUC of 0.91. A higher AUC indicates better performance of the model in identifying normal events from anomalous events with different threshold settings, hence specifying the strength, robustness and reliability of the model proposed in complex industrial environments.

TABLE 7. Detection time comparison on public transportation hub dataset.

Method	Detection Time (ms)
[2]	45
[6]	38
[18]	32
Proposed	25

The proposed model therefore represents a 25-millisecond detection time for real-time applicationst [Table 7], which is an important metric. Methods [2] take 45 milliseconds, [6], 38 milliseconds and [18], 32 milliseconds. Since the detection time will be reduced in the proposed model, anomalies will thus be identified within due time, which goes to ensure fast response and mitigation in dynamic and high-traffic environments. In other words, the results of many experiments carried out on several datasets validate a better performance by the proposed multimodal anomaly detection model. Substantial gains in accuracy, precision, recall, F1-score, AUC and time to detect, in comparison with prior methods, confirm that multimodal data fusion using deep learning techniques can permit efficient and robust anomaly detection in various surveillance environments. Next, we validate these results using ablation analysis for the proposed model in different scenarios.

A. VALIDATION & STUDY WITH ABLATION ANALYSIS

It is for this reason that the present study proposes a context-driven, multilevel dataset-based quantitative effectiveness review of the proposed multimodal anomaly detection model. For instance, consider a scenario where a surveillance system is set up in an urban environment [41]. For example, RGB images, infrared images and depth data are input data with key features and indicators extracted from each modality. In this study, the working of many components, including MDBM, MVAE, Attention-based Fusion Networks, DCA, ALI, DIM, LSTM, TCN and Transformer Networks with Temporal Encoding [Table 14], were analyzed and their results compared. The final outputs strongly prove that the model proposed in the results can detect the anomaly accurately.

Sample Values & Other Data Samples:

- RGB Image Features: Pixel intensity values, color histograms
- Infrared Image Features: Thermal intensity values, heat maps
- Depth Data Features: Distance measurements, 3D spatial coordinates
- Indicators: Movement patterns, temperature anomalies, object detection

TABLE 8. MDBM performance.

Feature	Method [2] Value	Method [6] Value	Method [18] Value	Proposed Value
Accuracy (%)	85.0	87.5	89.0	94.0
Precision (%)	82.0	84.5	86.5	92.0
Recall (%)	81.5	83.5	85.0	91.0
F1-Score (%)	81.8	84.0	85.7	91.5
AUC	0.85	0.87	0.89	0.94

TABLE 9. MVAE performance.

Feature	Method [2] Value	Method [6] Value	Method [18] Value	Proposed Value
Accuracy (%)	86.0	88.0	89.5	95.0
Precision (%)	84.0	85.5	87.0	93.0
Recall (%)	83.0	84.5	86.0	92.5
F1-Score (%)	83.5	85.0	86.5	92.8
AUC	0.86	0.88	0.90	0.95

MDBM improves all performance [Table 8] metrics as against the benchmark methods. Specifically, the proposed model has obtained an accuracy of 94.0%, a precision of 92.0%, a recall of 91.0%, an F1-score of 91.5% and an Area Under Curve of 0.94. These results demonstrate quite clearly the applicability of MDBM in learning joint representation from heterogeneous data sources for robust anomaly detection. [Table 9] The results obtained for the MVAE are accuracy 95.0%, precision 93.0%, recall 92.5%, F1-score 92.8% and AUC 0.95. This is because the MVAE learns better representations for capturing data distribution in multiple modalities and generates robust features that help boost the performance in anomaly detection.

TABLE 10. Attention-based fusion networks performance.

Feature	Method [2] Value	Method [6] Value	Method [18] Value	Proposed Value
Accuracy (%)	87.0	89.0	90.5	96.0
Precision (%)	85.0	87.0	88.5	94.0
Recall (%)	84.0	86.0	87.5	93.5
F1-Score (%)	84.5	86.5	88.0	93.8
AUC	0.87	0.89	0.91	0.96

[Table 10] The Attention-based Fusion Networks outperform the benchmark methods with an accuracy of 96.0%, precision of 94.0%, recall of 93.5%, F1-score of 93.8% and an AUC of 0.96. Leveraging attention mechanisms to highlight significant features from each modality proves highly effective in enhancing anomaly detection performance.

TABLE 11. DCA performance.

Feature	Method [2] Value	Method [6] Value	Method [18] Value	Proposed Value
Reconstruction Error	0.18	0.15	0.12	0.08
Latent Space Dimensionality	128	128	128	128
Normal Event Reconstruction	80%	85%	88%	96%
Anomaly Detection Precision	82%	86%	89%	94%

The [Table 11] Deep Convolutional Autoencoder (DCA) shows reduced reconstruction error (0.08) and improved normal event reconstruction accuracy (96%), significantly enhancing the precision of anomaly detection at 94%. This illustrates DCA's effectiveness in learning compact representations of normal patterns.

TABLE 12. ALI performance.

Feature	Method [2] Value	Method [6] Value	Method [18] Value	Proposed Value
Latent Representation Robustness	Medium	Medium	High	Very High
Adversarial Loss (Discriminator)	0.25	0.20	0.15	0.10
Anomaly Detection Precision	83%	87%	90%	95%
Anomaly Detection Recall	81%	85%	88%	94%

[Table 12] Adversarially Learned Inference (ALI) significantly improves the robustness of latent representations, with the adversarial loss of the discriminator minimized to 0.10. This results in high anomaly detection precision (95%) and recall (94%), showcasing ALI's efficacy in enhancing the model's robustness.

[Table 13] Deep InfoMax (DIM) achieves the highest mutual information maximization at 0.90, resulting in very high discriminability of the latent space. Consequently, DIM

TABLE 13. DIM performance.

Feature	Method [2] Value	Method [6] Value	Method [18] Value	Proposed Value
Mutual Information Maximized	0.70	0.75	0.80	0.90
Latent Space Discrim-inability	Medium	High	High	Very High
Anomaly Detection Precision	84%	88%	91%	96%
Anomaly Detection Recall	82%	86%	89%	95%

shows superior anomaly detection precision (96%) and recall (95%), underscoring its capability to learn informative and discriminative representations.

The precision and recall for the proposed model with its LSTM in modeling the temporal context are 92% and 91%, respectively; for TCN, 93% recall and 92% precision; and for the highest compared group, Transformer Networks, which achieve 95% recall and 94% precision. The results show the effectiveness of the proposed methods in capturing temporal dependences and improving the applied anomaly detection. In summary, This comprehensive benchmarking of the proposed model for multimodal anomaly detection against multiple context datasets indicates that the proposed method is clearly superior in performance. Such large improvements in accuracy, precision, recall, F1-score, AUC and speed of detection prove proper multimodal data utilization and the use of advanced deep learning approaches in a model. This is a robust and effective approach to anomaly detection that can, in fact, be used to enhance security in different scenarios and to increase video surveillance capabilities.

B. REAL-TIME ADAPTATION

The scope of these efforts is towards reducing the computational complexity and improving processing efficiency so as to make the model suited for real-time applications. Some promising directions in this regard could be pruning and quantization techniques which cut down on the number of parameters and the precision of weights that occur in a model without too significant loss in performance. This would enable the model to run on resource-constrained devices such as edge devices deployed in real-world surveillance networks, yet have much faster inference times while retaining high accuracy. More broadly, various strategies for distributed computing might be pursued; that is, distribute the processing load among multiple devices or servers. The model can be divided into smaller units, which may be processed in parallel or in sequence over a distributed network to reduce latency and enable the system to accommodate several camera feeds on real-time sets.

The model can further be optimized for real time execution through the use of streaming data processing frameworks. As opposed to processing whole batches of data sequentially, the system could process small, real-time windows of video frames and apply the associated temporal anomaly detection algorithms incrementally as new data become available. This would let the system respond immediately to anomalies and thus reduce the global reaction time in such dynamic environments like public transportation hubs, industrial sites, or crowded urban streets.

The real-world deployment of street surveillance in urban areas, as well as industrial site monitoring and public transportation stations, encounter the main challenge of heterogeneous and dynamic environments. Single-modality systems strictly rely on RGB data, whereas they may struggle with certain incidents such as low illumination, occlusions, or interactions of complexity. Here, this multimodal approach primarily relies on the combination of RGB, infrared and depth data to overcome any limitations related to a specific modality. For example, in low-light operations where RGB data fails, infrared data will capture thermal signatures; while depth data will yield spatial awareness. This all-inclusive approach makes the model highly relevant for real-world security systems, where conditions are unpredictable and environmental variations cross over accuracy in anomaly detection.

Additionally, direct application of the attention mechanism further enhances performance in real-world usage by focusing more sharply on the most pertinent information in the scene. In the real world, surveillance systems can introduce noisy or irrelevant information, such as pedestrians walking in the background or non-controlled lighting changes. The focus of such a fusion network will enable the model dynamically to pay more attention to salient features, such as suspicious behavior or unusual movement patterns, so that the system will continue to be correct even with very intense and cluttered environments. The particular use in crowded places, such as at a public event or transportation hub, can mean having to focus on critical anomalies in real time for effective security. Finally, the incorporation of temporal modeling techniques, such as LSTM, TCN and Transformer Networks, addresses the need for time unfolding anomaly detection. In industrial safety monitoring or traffic surveillance, some anomalies are equipped with equipment malfunction or aberrant driving behaviors that develop gradually rather than happening in one frame. This is a considerable advantage as the system captures long-term dependencies in the data. For example, an industrial system detecting that a machine is slowly overheating over a sequence of frames may turn on a pre-failure warning. In the analysis of traffic patterns, having sequences of erratic behavior over time rather than drawing conclusions from single instances of behavior allows better determination of potentially dangerous conditions.

Such techniques clearly elaborate how they are tailored in order to solve specific real world challenges; therefore this paper can indicate that it is not only technically robust but practically applicable towards the betterment of surveillance systems from almost all the different kinds of domains. It is due to this very relevance that the proposed model happens to be valuable in terms of its deployment in deploying the real world security safety and anomaly detection tasks.

TABLE 14. LSTM, TCN, transformer networks performance.

Method	LSTM Recall	LSTM Precision	TCN Recall	TCN Precision	Transformer Recall	Transformer Precision
[2]	82%	80%	83%	82%	85%	84%
[6]	85%	83%	86%	85%	88%	87%
[18]	87%	85%	88%	87%	90%	89%
Proposed	92%	91%	93%	92%	95%	94%

C. INCORPORATION OF OTHER DATA MODALITIES

Combining these existing RGB, infrared and depth modalities with audio data and text data would make the model robust for identifying subtle as well as complex anomalies. Audio data particularly helps in spaces where sensing alone based on vision is not adequate enough to identify some relevant activities, such as aggressive behavior, loud voices, or gunshots. This could inject an additional audio stream to process the pattern of sounds and correlate them with the visual information to generate a more holistic and contextual anomaly detection framework. For example, suspicious movements viewed through video data correlated with deafening, unusually loud noises picked up through audio can be considered as better proof that an incident is happening at a public place such as a subway. This may result in faster and perhaps much more accurate responses.

That would introduce another dimension of context into anomaly detection by leveraging textual data such as metadata from IoT devices, access logs, or social media streams. It may be applied in the industry to combine visual footage with textual data from equipment logs, for instance, towards more accurate and pre-escalation detection of operational anomalies such as those due to equipment malfunction. Another way is that NLP techniques can be utilized to examine real-time messages or alerts that might be generated by security personnel or automated systems. So, the model can make associations between visual anomalies and textual reports for an all-around understanding of the situation.

The proposed approach then could be a versatile and highly effective tool within a wide range of surveillance applications, from urban security to industrial safety monitoring, if it is successfully scaled up in integrating the additional modalities into the model and optimizing the model for real-time operation. Then, new research avenues will be opened for more adaptive, context-aware and efficient systems of anomaly detection that can address the challenging problems mentioned above in present surveillance environments.

D. COMPARISON WITH THE BASELINE MODELS

The proposed multimodal fusion model was compared against the following three commonly used baseline models for anomaly detection: 1 CNN on RGB alone; 2 AE for unsupervised learning on single-modality data; and 3 RNN, LSTMs in particular, for temporal pattern detection using RGB video data. Each of the above models was chosen as a baseline due to its wide adoption and effectiveness in traditional anomaly detection tasks.

Comparing with a standard CNN that relies solely on RGB information, the multimodal fusion model resulted in an improved accuracy of about 12% from around 83% up to 95%. Generally, the RGB-only model suffers badly in low-light environments or scenes containing occlusions where its performance degrades seriously. Unlike this, anomalies that seemed invisible in the RGB feed were detected by integrating infrared and depth data. This thus reveals how multimodal data integration enable the proposed model to work better at anomaly detection under diverse environmental conditions where an incomprehensible information set is created by a single modality.

Unlike an autoencoder (AE) model that aimed at single-modality unsupervised anomaly detection, the proposed model is with better performance because it was able to learn robust features across different modalities. The false positive rate was about 15% higher in the AE model than the proposed approach. This is because the AE is strictly confined to learning the distribution of a single source of data and cannot capture complex relationships that may exist between multiple modalities like RGB, infrared and depth. The MVAE, in the proposed model, learns interdependencies between multiple modalities and decreases false positives by better identification of normal patterns and recognition of anomalies that evolve over various types of data.

Last but not least, the proposed method achieves a 15% improvement in recall as compared to Recurrent Neural Networks (RNN) and the LSTM-based models that capture temporal patterns in video sequences and exhibits superior long-term anomaly detection accuracy. Although the LSTM model performs efficiently with a temporal dependency learning function, a highly dependent model on a single data modality - the RGB video-limits performance to subtle thermal or spatial changes that may not be detected solely by visual data. The proposed model's use of Temporal Convolutional Networks and Transformer Networks with self-attention mechanisms offers a more flexible approach in terms of handling the temporal modeling, particularly when multimodal data is provided. The self-attention mechanism offered by the Transformer has the ability to provide a weighting on the feature importance over time, further improving the ability to detect long-duration or evolving anomalies. Limitation of Baseline Models and Proposed Method.

However, as the proposed approach would undoubtedly trounce the baselines, one needs to say that each method, including the proposed approach, is weak in some ways. For example, baseline CNN models on RGB data perform especially well in open, well-lit environments but break down in low-light or complex situations where infrared or depth information would be advantageous. Similarly, Autoencoders is good at detecting anomalies in particular types of data and not so generalization-friendly if applied to a variety of multimodal data sources. LSTM-based RNN models are highly based on the quality and availability of data with temporal information and sensitive to missing critical visual cues especially in adverse environments.

The proposed multimodal fusion model overcomes many of the challenges pointed out above but in turn is very computational intensive and computationally complex. Combining multiple modalities and even advanced architectures like MDBM or attention-based fusion networks are highly computational intensive that require significant resources, which would not be a possibility in real-time surveillance applications that must operate in resource-constrained environments. This model, though is robust to noisy data, even this performance will suffer if there are several modalities simultaneously losing data or receiving severe noise at the same time, such as when sensor calibrations are poor or interference in the environment is significant.

E. WEAKNESSES OF THE PROPOSED MODEL

The key weakness identified in the proposed model was computational complexity. Integration of multiple modalities and advanced models like MDBM, MVAE, or attention-based fusion networks would require a significant amount of computational power, which might severely limit real-time performance on a low-constrained device. Even if the model was to run efficiently on high-end hardware, for example, the NVIDIA Tesla V100 GPU, its deployment on an edge device or in the real-time surveillance system could be facilitated only by optimization, perhaps through techniques like pruning or quantization, to minimize the computational load. These approaches can reduce the number of parameters and speed up inference without significant accuracy loss.

This may also be a challenge in that the model is based on full multimodal data. In a real-life deployment, some sensors could fail or get noisy. The model's performance would degrade with such conditions. To handle this, future work can include developing more robust fusion techniques that adjust the importance of available modalities based on missing or corrupted data. Another direction is to apply transfer learning to improve generalization in different environments and reduce the need for many labeled training samples.

By involving such complex patterns of experimental setup, sensitivity analysis and limitations in the model, the work will be well-based and technically sound, reproducible in nature and emphasize the effectiveness of the proposed anomalous detection system useable in real-world applications.

VI. CONCLUSION AND FUTURE SCOPES

The proposed model works very well in the detection of anomalies, with an accuracy that was very high compared to other related methods it applied on CCTV videos for various surveillance contexts. It uses the following techniques: MDBM, MVAE, Attention-based Fusion Networks, DCA, ALI, DIM, LSTMs, TCNs and Transformer Networks with Temporal Encoding to link heterogeneous data sources efficiently with advanced learning techniques, improving anomaly detection performance. Results demonstrate that the proposed model achieves large gains in all the major performance indicators: The proposed model has achieved accuracy of 95.1%, in contrast to all existing state-of-the-art methods with an accuracy of 90.2% on the Urban Street Dataset alone for different scenarios. On the Industrial Site Dataset, the model has a precision of 94.3 percent, versus 89.5 percent for method, with an AUC of 0.96, much higher than 0.91 for that method alone. Besides, recall of the Public Transportation Hub Dataset also reaches 92.5 percent, which is 86.8 percent higher than that of method [18], while detection time is reduced to 25 milliseconds compared with the 32 milliseconds in method [18]. These results suggest that the proposed model can be more reliable and efficient for highly accurate anomaly detection under different complex scenarios.

A. LIMITATION MITIGATION AND FUTURE WORK

Future work may include some techniques, such as pruning and quantization of the proposed method, with the goal of parameter reduction, aiming to make the model more efficient so that it can be run on edge devices for real-time surveillance. Distributed architectures for computing can be one of the solutions that the model may process streams of data across multiple devices or servers in parallel, so that the decision time can be reduced. Future work could also be along the lines of further research in developing robust techniques to handle data, especially those using missing or noisy modalities by dynamically weighting the available modalities or using data imputation methods to reconstruct missing data from the remaining modalities.

The addition of new modalities, including audio and text information, could also lead to more advanced anomaly detection capabilities for the model. For example, fusion of auditory patterns with visual and thermal inputs can improve the detection capability of suspicious activities in public spaces. Also, textual data from industrial sensors or access control logs would help infer additional context regarding anomalies in industrial or security contexts.

B. FUTURE SCOPE

Although the proposed model comes out very good in performance, it opens many future avenues that can be taken to probe into its increased capability. One view could be the integration of further data modalities, such as audio and text data, which are able to provide richer context information

and let the model detect subtle and complex anomalies. Other promising directions include more advanced fusion techniques that adapt dynamically to differing data quality and availability, such as will help make models more robust in real-world conditions where the data from some modalities might either be missing or too noisy. A further successful area for improvements is in the development of richer temporal modeling techniques. For instance, it can be improved by recurrent neural networks with attention mechanisms or including graph-based models to better capture spatiotemporal dependencies for the detection of anomalies that evolve over temporal instance sets. On the other hand, transfer learning and domain adaptation methods are also available to make the model generalize quite well across a multitude of surveillance environments, just avoiding the requirement for large labeled training data samples. Furthermore, computational efficiency and scalability in deploying the proposed model for real-time surveillance systems is yet to be investigated. Some of such possible avenues are model pruning techniques and quantization, or distributed computing for establishing the effectiveness of the model under real-world surveillance deployments that mostly consist of resource-constraint conditions. Basically, the proposed approach of multimodal anomaly detection establishes state-of-the-art results, clearly outperforming prior work in accuracy, precision, recall, F1-score, AUC and detection delays. In this case, promising avenues for future work toward real-time anomaly detection in CCTV video analysis and more modalities in data would be advanced temporal modeling and computational efficiency levels.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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